

FORT DODGE, IA, USA

WMO: 725490

Lat: 42.550N

Lon: 94.183W

Elev: 352

StdP: 97.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94933

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-20.6	-27.5	0.3	-22.3	-24.6	0.4	-20.3	11.0	-3.9	10.0	-3.6	2.2	320	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.7	24.2	31.3	23.5	29.8	22.8	25.7	30.7	24.7	29.7	23.8	28.4	5.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.8	29.0	22.9	18.4	27.9	22.3	17.8	27.2	81.9	31.2	77.0	29.7	72.9	27.9	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	8.9	8.0	-27.3	35.2	2.9	1.8	-29.4	36.4	-31.1	37.5	-32.7	38.5	-34.8	39.8
			-27.6	27.6	2.7	1.0	-29.6	28.3	-31.2	28.8	-32.7	29.4	-34.7	30.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-7.7	-5.3	1.9	8.9	15.4	21.0	23.1	21.6	17.7	10.4	2.6	-4.7
	DBStd	12.06	7.18	7.14	6.83	5.70	4.74	3.61	3.25	3.05	4.70	5.40	6.02	6.61
	HDD10.0	2088	548	429	265	85	8	0	0	0	3	61	233	456
	HDD18.3	3912	806	663	511	286	116	15	3	7	67	252	473	714
	CDD10.0	1654	0	0	13	53	177	330	405	358	234	74	9	0
	CDD18.3	436	0	0	1	4	26	95	149	107	48	6	0	0
	CDH23.3	4064	0	0	5	64	269	894	1408	878	473	69	2	0
	CDH26.7	1328	0	0	0	16	73	300	512	266	144	17	0	0
Wind	WSAvg	3.6	3.7	3.9	4.2	4.6	4.1	3.4	2.8	2.6	3.1	3.7	3.9	3.7
Precipitation	PrecAvg	857	22	23	54	85	112	128	114	108	85	57	42	27
	PrecMax	1225	67	78	156	216	262	306	286	290	347	156	135	73
	PrecMin	555	1	4	1	15	43	46	28	12	9	4	0	1
	PrecStd	164	15	18	35	45	51	63	57	59	65	38	32	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	14.1	23.0	28.7	32.2	33.7	35.2	33.7	32.6	28.9	22.0	13.6
		MCWB	5.9	9.7	14.8	17.7	21.3	23.9	26.1	24.7	22.2	19.0	14.7	10.5
	2%	DB	6.3	10.0	18.8	24.9	28.6	32.1	32.8	31.7	30.9	25.0	17.6	8.9
		MCWB	3.4	7.2	13.0	15.5	19.5	22.9	25.3	24.7	22.3	16.7	12.3	6.2
	5%	DB	3.0	7.0	15.9	22.1	26.4	30.0	31.4	29.1	27.9	22.3	14.0	6.4
		MCWB	1.1	4.3	11.4	13.7	17.7	22.0	24.6	23.4	20.4	15.9	9.4	3.7
	10%	DB	2.0	3.8	12.2	18.8	23.9	27.9	29.1	27.7	26.2	19.0	12.0	3.7
		MCWB	0.6	1.9	8.4	12.2	16.5	20.6	23.4	22.2	19.3	13.9	8.2	1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	11.0	17.0	18.9	23.0	25.8	27.8	26.8	24.3	21.4	15.4	11.4
		MCDB	10.1	13.5	21.2	26.8	29.6	31.4	32.8	31.3	30.1	26.5	19.9	13.2
	2%	WB	3.7	7.2	14.1	16.5	20.8	24.3	26.3	25.3	23.0	18.6	12.7	6.8
		MCDB	6.1	9.8	18.2	22.7	27.2	29.6	31.1	30.2	28.3	23.3	16.6	8.7
	5%	WB	1.7	4.2	11.4	14.7	19.4	23.0	25.3	24.1	21.8	16.5	10.1	4.0
		MCDB	3.3	6.7	15.6	20.7	24.6	28.4	30.3	28.4	26.5	20.8	13.8	6.3
	10%	WB	0.2	2.0	8.2	12.6	17.7	21.8	24.2	23.1	20.4	14.5	7.9	1.9
		MCDB	1.5	3.9	11.6	18.0	22.4	26.9	28.9	27.0	24.9	19.0	11.1	3.9
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.3	10.1	11.9	11.5	11.1	10.9	10.7	12.5	11.7	10.1	8.6
		MCDBR	11.3	11.7	15.2	17.2	14.3	12.9	11.9	12.0	13.9	15.1	14.5	11.8
		MCWBR	9.2	9.4	10.1	9.8	7.4	6.3	6.3	6.6	7.1	8.7	9.7	9.4
	5% WB	MCDBR	10.8	11.2	14.2	14.4	12.1	11.1	10.8	10.8	11.8	12.5	13.0	11.2
		MCWBR	9.0	9.1	9.9	9.3	7.2	6.6	6.3	6.4	6.7	8.4	9.4	9.2
Clear-Sky Solar Irradiance	taub	0.293	0.288	0.317	0.369	0.396	0.418	0.426	0.413	0.383	0.339	0.296	0.277	
	taud	2.383	2.418	2.438	2.321	2.313	2.297	2.249	2.285	2.328	2.446	2.507	2.460	
	Ebn at Noon	847	921	931	898	878	856	845	844	845	847	846	842	
	Edh at Noon	79	90	100	122	127	129	134	125	112	88	70	67	
All-Sky Solar Radiation	RadAvg	1.88	2.91	3.79	4.75	5.60	6.21	6.42	5.45	4.46	2.99	2.02	1.52	
	RadStd	0.19	0.23	0.38	0.43	0.41	0.50	0.37	0.37	0.30	0.38	0.15	0.11	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page